

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 Prelim

H2 BIOLOGY

9744/3

Paper 3 (Mark Scheme)

Monday

17th September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

For Examiners' Use	
Section A	
1	/35
2	/7
3	/8
Section B	
4 or 5	/25
Total	/75

This document consists of **23** printed pages.

[Turn over

2
Section A

Answer all questions.

QUESTION 1

- (a) Potential cancer cells experience a series of alterations during oncogenic transformation that confers them new features. These features are both genetic and phenotypic in nature. One prominent feature is the reprogramming of their energy metabolism. This phenomenon has been termed 'The Warburg Effect'.

The Warburg Effect describes the observation that cancer cells exhibit an enhanced glycolysis without undergoing oxidative phosphorylation, even when oxygen is abundant. This leads to another prominent feature of cancer – the acidification of the extracellular environment of tumours due to the secretion of an overproduced molecule by cancer cells as a result of enhanced glycolysis.

Since the observation, many journal reports have surfaced to provide explanations to the Warburg Effect. Some of these include various mutations, upregulation or downregulation of expression of various genes involved in respiration, for example, those that code for the enzyme ATP synthase.

- (i) Outline the 'series of alterations' that could occur in the potential cancer cells.

.....[5]

- 1 results in loss of arrest of cell division + loss of ability for DNA repair + loss of apoptosis;
- 2 Gain of function mutation of proto-oncogene;
- 3 lead to over-stimulation of the cell cycle / cell keeps dividing;

(max 2 AVP: ref. gene amplification of proto-oncogene; ref. translocation of proto-oncogene to be under control of more active promoter / translocation of active promoter to upstream of proto-oncogene;)

(max 3)

(This allows for subsequent accumulation of many other mutations)

- 4 Some mutations may cause activation of telomerase gene; (Reject: mutation of telomerase gene)
- 5 Some mutations cause a loss of ability to differentiate;
- 6 Some mutations cause cells to no longer exhibit anchorage dependence / loss of cell adhesion;
- 7 Some mutations cause a loss of contact inhibition / density-dependence (and cells do not stop dividing);
- 8 Mutations can also lead to angiogenesis / formation of new network of blood vessels to the cancer cells;
- 9 Some mutations allow metastasis to occur/ cancer cells are able to break loose and travel in the bloodstream and invade other tissues to form secondary tumors;
- 10 AVP; (e.g. elaboration of preamble without direct copying e.g. mutations in / upregulation / downregulation of genes involved in **respiration**) (Reject: ATP synthase as it is in the preamble)

- (ii) Explain how an enhanced glycolysis can lead to the acidification of the tumours' extracellular environment.

-[2]
- 1 **Lactate / lactic acid** produced from lactate fermentation;
 - 2 to **regenerate NAD⁺** from NADH for aerobic glycolysis to continue;
 - 3 ref. secreted out of cell into extracellular environment; (Reject: diffusion)
(Accept: H⁺ ion being pumped out)

The fact that most cancer cells exhibit the Warburg Effect has been used clinically as a basis for cancer diagnosis in patients. This method of diagnosis is known as PET scanning as outlined below:

- Patient is first injected with ¹⁸F-fluorodeoxyglucose, a radioactive glucose analogue
- ¹⁸F-fluorodeoxyglucose can be taken up by respiring cells
- In the cytoplasm, ¹⁸F-fluorodeoxyglucose can be phosphorylated by hexokinase but cannot be metabolised further in glycolysis
- Presence of radioactivity is determined in the various sites of the body

- (iii) Predict the observation in PET scanning on cancer cells compared to non-cancer cells.

-[1]
- 1 Ref. **higher radioactivity** in cancer cells compared to non-cancer cells; (Reject: presence of radioactivity as there must be a comparison)

- (iv) Explain the prediction made in (iii).

-[2]
- 1 Cancer cells exhibit **enhanced glycolysis** / undergo glycolysis without oxidative phosphorylation;
 - 2 Leads to **higher uptake/accumulation** of **¹⁸F-fluorodeoxyglucose**; (and hence radioactivity in cancer cells as ¹⁸F-fluorodeoxyglucose is not metabolised further)

- (v) Suggest why cancer cells require more glucose than non-cancer cells to meet energy demands.

-[1]
- 1 As (enhanced) glycolysis synthesises only **net 2 ATP** molecules as compared to oxidative phosphorylation that synthesises **32 / 38 ATP** molecules **per glucose** molecule;
 - 2 Ref. cancer cells may also need more **ATP** molecules as it is **dividing uncontrollably**;

- (b) Central to the energy metabolism of aerobically respiring cells is the enzyme ATP synthase. It consists of multiple subunits, coded for by several genes.

The enzyme is located on the inner mitochondrial membrane and catalyses the reversible reaction of ATP synthesis from ADP in intact mitochondria. Here, the **proton motive force** is required to drive the ATP synthesis.

- (i) Explain how the structure, including protein components, of the inner mitochondrial membrane is significant in driving the reaction of ATP synthase towards ATP synthesis during aerobic respiration.

[4]

[Important structure – hydrophobic fatty acid tails]

- 1 **Hydrophobic** fatty acid tails / hydrocarbon chains / hydrophobic core of inner mitochondrial membrane **repels / does not allow hydrophilic / charged H^+ ions** to pass through membrane;
- 2 Allows **proton gradient / proton motive force** to be established;

[Important composition – series of electron carriers]

- 3 Electrons passed down a series of **electron carriers / ETC** present on membrane with increasing electronegativity and in order of **decreasing energy levels**; (until they reach final electron acceptor – oxygen)
- 4 Energy released during transfer of electrons along series of electron carriers used to **pump H^+ ions from mitochondrial matrix into intermembrane space**;

[Important composition – ATP synthase]

- 5 **ATP synthase** embedded in an orientation that allows facilitated **diffusion of H^+ ions from intermembrane space to mitochondrial matrix** to be coupled with ATP synthesis;

In a cell-free or mitochondrial-free system, the same enzyme can also catalyse ATP hydrolysis when ATP is present. This reaction can be exploited for the investigation of ATP synthase activity.

To examine the enzymatic activity of ATP synthase, the indirect method of measuring the relative NADH concentration in the solution can be utilised as shown below in **Fig. 1.1**.

ATP produced when phosphoenolpyruvate is converted to pyruvate by pyruvate kinase, is acted upon by ATP synthase to form ADP.

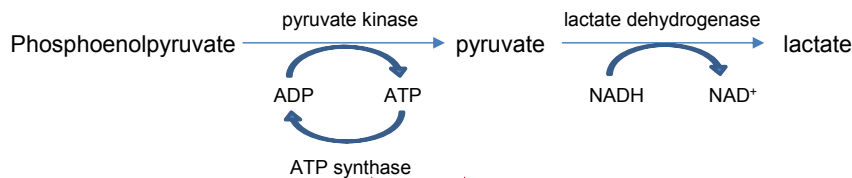


Fig. 1.1

Commented [TYMY1]: Change diagram labels to Fig 1.1 Fig 1.2

The relative concentration of NADH can be tracked with a spectrophotometer as NADH absorbs light at 340nm while NAD^+ and other molecules in the system do not.

- (ii) In examining the activity of ATP synthase using the reactions in **Fig. 1.1**, explain why phosphoenolpyruvate, ATP and NADH need to be in high concentrations.

.....[2]

- 1 So that phosphoenolpyruvate, ATP and NADH are **not limiting** factors in the overall reaction;
- 2 Ref. to allow reaction to proceed forward;
- 3 Allows rate of **depletion of NADH / decrease in absorbance** to reflect / to be the same as **ATP synthase activity**;

The reactions in **Fig. 1.1** can also be used to determine the effect of a particular drug on ATP synthase. Curcumin, a phytochemical isolated from the rhizome of turmeric, has been shown to affect the activity of ATP synthase. However, its effect varies according to the source of ATP synthase.

Fig. 1.2 and **Fig. 1.3** show the effect of curcumin on ATP synthase molecules isolated from the liver mitochondria and brain mitochondria of the same rat respectively.

% control refers to ATP synthase activity with reference to activity at 0 μM curcumin.

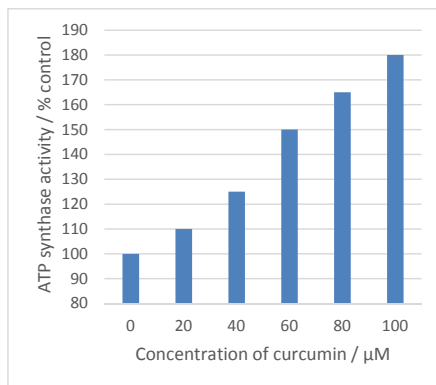


Fig. 1.2: Effect of curcumin on ATP synthase isolated from rat liver mitochondria

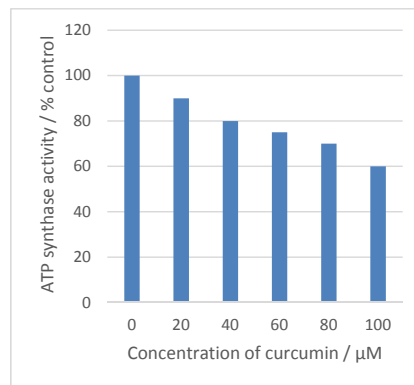


Fig. 1.3: Effect of curcumin on ATP synthase isolated from rat brain mitochondria

- (iii) With reference to **Fig. 1.2**, describe the effect of curcumin on rat liver mitochondrial ATP synthase.

.....[2]

- 1 **[Curcumin effect in general]** Curcumin **activates / increases activity** of ATP synthase;
- 2 **[Quote data]** ATP synthase activity is above 100% control for all concentrations of curcumin OR quote for any single curcumin concentration;
OR
- 3 **[Trend]** As curcumin concentration increases, ATP synthase activity increases;
- 4 **[Quote data]** e.g. as curcumin concentration increases from 0/20 μM to 100 μM , ATP synthase activity increases from 100/110% control to 180% control;

- (iv) Suggest one process that can occur within cells, and explain how it can lead to the differential effects of curcumin on ATP synthase molecules isolated from liver and brain mitochondria of the same rat.

.....[4]

- 1 **[Process 1] Different genes** that code for the subunits are **expressed** in liver and brain cells;
- 2 Ref. each subunit has a different **3D conformation / tertiary structure**;
- 3 Ref. ATP synthase in liver and brain mitochondria consisting of **different subunits** and **quarternary structure**
- 4 Both ATP synthase still complementary to curcumin, but curcumin **binds differently** to both, leading to differential effect; (Reject: curcumin binding to active site)

OR

- 1 **[Process 2] Alternative splicing** of ATP synthase subunit pre-**mRNA** from a particular gene (in liver and brain cells);
- 2 Leading to subunits with different **3D conformation / tertiary structure**;
- 3 Thus, ATP synthase in liver and brain mitochondria will have different **quarternary structure**;
- 4 Both ATP synthase still complementary to curcumin, but curcumin **binds differently** to both, leading to differential effect; (Reject: curcumin binding to active site)

OR

- 1 **[Process 3] Different post-translational modification** of ATP synthase subunit in liver and brain cells;
- 2 Leading to subunits with different **3D conformation**;
- 3 Thus, ATP synthase in liver and brain mitochondria will have different **quarternary structure**;
- 4 Both ATP synthase still complementary to curcumin, but curcumin **binds differently** to both, leading to differential effect; (Reject: curcumin binding to active site)

OR

- 1 **[Process 4] Mutation(s) in any genes** that code for the subunits in liver or brain cells;
- 2 Ref. the **corresponding subunits** in liver or brain cells having different primary structure and hence **3D conformation / tertiary structure**;
- 3 Thus, ATP synthase in liver and brain mitochondria will have different **quarternary structure**;
- 4 Both ATP synthase still complementary to curcumin, but curcumin **binds differently** to both, leading to differential effect; (Reject: curcumin binding to active site)

In many prokaryotes, the multi-subunit ATP synthase expression is under the control of a single operon.

(v) Suggest how the ATP synthase operon is organised.

.....[2]

- 1 **Structural genes** that code for the **ATP synthase subunits** clustered together;
- 2 Under the control of a single **promoter**;

- (c) ATP synthase deficiency is a disorder in which individuals are deficient in the enzyme, leading to a wide variety of signs and symptoms affecting many organs and systems of the body. The disorder can range from being mild to life-threatening.

Different mutations in different nuclear genes can result in the deficiency. These can generally be grouped into 2 categories. The first being mutations at gene loci coding for an ATP synthase subunit, for example, *ATP5E*.

The second being mutations at gene loci coding for a protein that is required for the proper assembly of the ATP synthase subunits to form the functional enzyme, for example, *TMEM70*.

- (i) Justify the claim that the severity of the disorder in an individual depends on his/her genetic makeup.

-[3]
- 1 **Different mutations** in a gene coding for an ATP synthase subunit / e.g. *ATP5E* lead to **different efficiency of ATP synthase**;
 - 2 **Different mutations** in a gene coding for a protein that is required for the proper assembly of the ATP synthase subunits to form the functional enzyme / e.g. *TMEM70* lead to **different efficiency of ATP synthase / different proportion of functional ATP synthase**;
 - 3 Ref. different effects of gene mutations (e.g. insertion/deletion not in multiples of 3 leading to frameshift vs substitution leading to silent/neutral mutation);
 - 4 Ref. **different combinations** of mutations at different loci;
 - 5 Ref. **additive effect** of mutations at multiple loci;
 - 6 Lead to **different ATP synthesis efficiency / ability** in different patients and hence different severity of the disorder;

ATP synthase deficiency cases have been reported frequently in the Roma population, an ethnic group living mostly in Europe. Various journal articles have collectively reported the following:

- *TMEM70* gene defect was identified as a novel cause of autosomal recessive ATP synthase deficiency in 2009
- Most of the patients with *TMEM70* gene mutations share a common Roma descent
- In all genotyped cases for patients of Roma origin, an adenine to guanine substitution was found, located in the splicing site of intron 2, which leads to aberrant splicing and thus preventing synthesis of the functional protein
- If the disorder is present as a result of this substitution mutation, a fatal outcome or life-threatening symptoms are expected

References:

- 1) Anne K. Braczynski, Stefan Vlaho, Klaus Müller, et al. (2015). ATP Synthase Deficiency due to *TMEM70* Mutation Leads to Ultrastructural Mitochondrial Degeneration and Is Amenable to Treatment. *BioMed Research International*, vol. 2015, Article ID 462592, 10 pages. <https://doi.org/10.1155/2015/462592>.
- 2) Josef Houštěk et al. (2009). *TMEM70* protein — A novel ancillary factor of mammalian ATP synthase. *Biochimica et Biophysica Acta – Bioenergetics*. 1787(5): 529-532.
- 3) Spiegel R et al. (2011). *TMEM70* mutations are a common cause of nuclear encoded ATP synthase assembly defect: further delineation of a new syndrome. *J Med Genet*. 48(3):177-82.

- (ii) It can thus be deduced that a particular mutation is associated with people of Roma origin that can result in ATP synthase deficiency.

With reference to the information given, suggest how the mutation could have been preserved in present day people of Roma descent despite leading to fatal outcomes.

-[3]
- 1 Mutation is **recessive**;
 - 2 Ref. effect of mutation can be **masked** by a dominant allele coding for a **functional TMEM70 protein**;
 - 3 **Heterozygous** individuals with the recessive mutation able to **survive and reproduce**, thus, the recessive allele can be passed down to the offspring, and can be preserved in the population;

ATP synthase deficiency can also result from mutations in the mitochondrial genes, for example, *ATP6*. The *ATP6* gene codes for a subunit of ATP synthase.

Fig. 1.4 shows a pedigree chart where individual 2, who is female, carries a mutation in the *ATP6* gene and thus suffer from the disorder. Individual 1 does not carry any mutations in the *ATP6* gene. Both individuals 1 and 2 do not carry any other mutations that can result in the disorder.

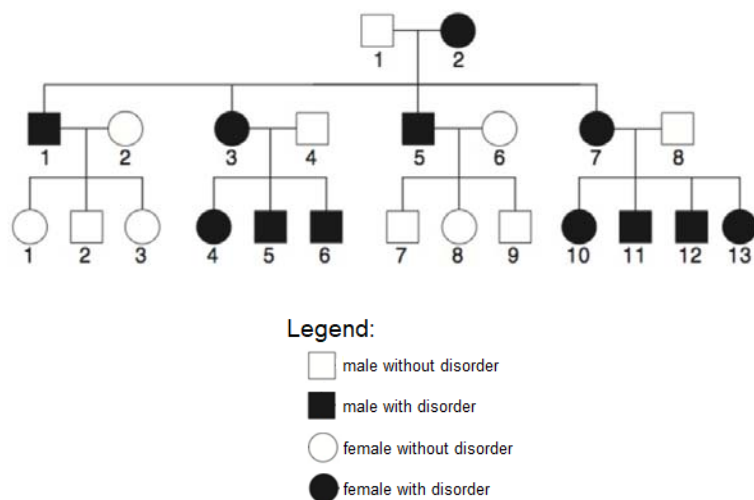


Fig. 1.4

(iii) With reference to **Fig. 1.4**, explain how the disorder is inherited.

.....[4]

- 1 Inheritance is via **maternal line** / mothers;
- 2 **All children**, male or female, will inherit the disorder from the **mother**;
- 3 **[Quote data]** e.g. All children individuals 4, 5 and 6 (generation 3) of individual 3 will have the disorder OR All children individuals 10, 11, 12 and 13 (generation 3) of individual 7 will have the disorder OR All children individuals 1, 3, 5 and 7 (generation 2) of individual 2 will have the disorder
- 4 Zygote obtains all **mitochondria** from **mother** and none from father;
- 5 **ATP6** gene present only in mitochondrial DNA and **not present in nuclear DNA/chromosomes**;

QUESTION 2

Climate change, in the form of global warming, is expected to have an impact on various organisms. This impact however, varies geographically, particularly for insects. **Fig. 2.1** and **Fig. 2.2** below show the fitness curves of representative insects from temperate and tropical locations respectively.

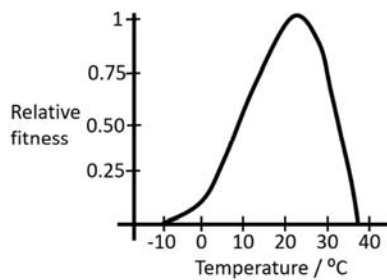


Fig. 2.1: Fitness curve of representative insect from temperate location

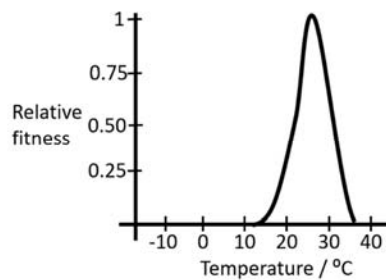


Fig. 2.2: Fitness curve of representative insect from tropical location

The mean annual temperature of the temperate location is 11 °C, with a typical temperature range from 1 to 20 °C. On the other hand, the mean annual temperature of the tropical location is 27 °C, with a typical temperature range from 21 to 31 °C.

(a) Suggest what is meant by the term relative fitness in **Fig. 2.1** and **Fig. 2.2**.

-[2]
- 1 a measure of evolutionary success as indicated by the number of surviving offspring left to produce the next generation;
 - 2 [reference to / as compared to **fitness at optimum temperature** (relative fitness of 1);

Commented [TYMY2]: Change diagram labels to Fig. 2.1, Fig 2.2 etc

Same for Q1 Diagrams

Commented [TYMY3]: change

- (b) With reference to **Fig. 2.1** and **Fig. 2.2** and the information given, predict and explain which insects, from the temperate or tropical location, would face a greater extinction risk as a result of global warming, assuming the warming elevates temperatures equally at both locations.

Commented [TYMY4]: change

.....[4]

- 1 Insects from the **tropical location**;
- 2 As these insects are already living **close to / just under / around /** (sometimes) **beyond** their **optimum temperature** for maximum relative fitness of 1 / these insects have a **narrower temperature tolerance**;
- 3 [Quote] Optimum temperature = 26 - 28°C (Accept: any figure within range) + mean annual temperature of the tropical location is 27°C OR typical temperature range from 21 to 31°C;
- 4 Ref. temperature predicted to increase by 2 – 3 °C as a result of global warming;
- 5 Ref. lower relative fitness beyond optimum temperature / Ref. temperature increase likely to lower relative fitness, thus, increasing extinction risk;
- 6 Ref. temperature increase likely to increase relative fitness of insects from temperate location instead, reducing extinction risk;

- (c) Suggest one strategy that the insects in (b) can employ to reduce the impact of global warming on themselves.

.....[1]

- 1 Migration to cooler climate / regions / habitats / higher altitude / latitude;

QUESTION 3

- (a) Proper activation of CD4 T cells is essential for an effective humoral immune response. This activation requires the involvement of antigen presenting cells, such as dendritic cells, that provide signals for the activation.

One such important signal is the presence of co-stimulatory molecules, such as CD80 or CD86, on the dendritic cell surface. These molecules are expressed and upregulated in dendritic cells via the Toll-like receptor signalling pathway. Toll-like receptors belong to a group of receptors known as Pattern Recognition Receptors that recognise typical surface molecules of pathogens as ligands.

CD4 T cell activation can occur via the natural route during infection or artificially via vaccination. Vaccines, when formulated effectively, could provide long term protection against a particular disease. Modern day vaccines that utilise only purified antigens generally do not evoke a strong immune response as compared to older style vaccines utilising live or killed whole bacteria.

- (i) Outline how activation of CD4 T cells is important for an effective humoral immune response.

.....[4]

- 1 Activated CD4 T cells give rise to many clones of activated **T helper cells** that are required for the activation of B cells;
- 2 Activated **T helper cells** have receptors that bind to (complementary / specific) **antigen** fragment displayed on a **class II MHC molecule** on the **B cell**;
- 3 Activated **T helper cells** release **cytokines**;
- 4 Activated B cells then **proliferate** and **differentiate** into **plasma cells** and memory B cells / undergoes **clonal selection** and **clonal expansion** to form **plasma cells**;
- 5 Plasma cells secrete **antibodies**;

- (ii) Explain how vaccines can provide a long term protection against a disease.

.....[2]

- 1 Ref. vaccines consist of antigen + example purified antigens / live or killed whole bacteria (preamble)
- 2 Ref. to immunological **memory** (e.g. Once activated, B and T cells undergo multiple cell divisions to produce a population of cells identical to the original cell where some cells in the clone become **memory cells**);
- 3 Memory cells are **long-lived cells** that can give rise to **effector cells** if the same antigen is encountered later in life / Encounter with the same pathogen again from which the vaccine was derived, during a real infection, triggers a **rapid** and strong **secondary immune response**;

- (iii) For vaccines utilising purified antigens to be as effective as older style vaccines utilising live or killed whole bacteria, adjuvants need to be added to these vaccines.

Based on the information given, suggest what can be used as an adjuvant and explain how it contributes to the vaccine's effectiveness.

-[2]
- 1 Bacterial cell wall (components) / flagella / Lipopolysaccharides / any named surface component of bacteria / ligands that bind to Toll-like receptor;
max 1
 - 2 Leads to **expression / upregulation of co-stimulatory molecules / CD80 / CD86** on the **dendritic cell / APC surface**;
 - 3 for **proper activation of CD4 T cells**;

Section B

Answer **one** question only.

QUESTION 4

- (a) Cellulose is the most abundant biopolymer on earth. It forms a significant proportion of the dry mass of plants.

Describe the Calvin cycle and explain the origin of the carbon atoms in cellulose, taking into consideration the number of molecules involved. [12]

Phase 1: Carbon dioxide fixation

1. **CO₂** (1-carbon molecule) **enters** the Calvin cycle, and is **fixed by combining** with a 5C compound called **ribulose-1,5-bisphosphate / RuBP**;
2. catalysed by the enzyme **rubisco / RuBP carboxylase**;
3. to form an **unstable intermediate 6C compound**;
4. The unstable 6C compound immediately splits into half to form **2 molecules** of a 3C compound called **glycerate-3-phosphate / GP / PGA**;

Phase 2: glycerate-3-phosphate (GP) reduction

5. **GP / PGA** is reduced to **triose phosphate / glyceraldehyde-3-phosphate / TP / PGAL / G3P**;
6. The electrons (hydrogen) for this **reduction** come from **NADPH**;
7. and the **energy** for this step comes from **ATP**;
8. ref. NADPH and ATP produced from the **light-dependent reactions**
9. For every **3** molecules of **CO₂**, there are **6** molecules of **triose phosphate / glyceraldehyde-3-phosphate / TP / PGAL / G3P**;
10. but only **1** molecule of **triose phosphate / glyceraldehyde-3-phosphate / TP / PGAL / G3P** (3C) **exits** the cycle, and will be used by the plant cell to synthesise carbohydrate like glucose (sugar);

Phase 3: Ribulose bisphosphate (RuBP) regeneration

11. The other **5** molecules of **triose phosphate / glyceraldehyde-3-phosphate / TP / PGAL / G3P** must be **recycled to regenerate 3** molecules of **RuBP**;
12. 3 more molecules of **ATP** from light dependent reaction used;

Max 8 pts

[explain origin of carbon atoms in cellulose]

13. Origin: **Carbon dioxide**;
14. Cellulose is a polymer of **β-glucose**;
15. Each β-glucose comes from **2** molecules of **triose phosphate / glyceraldehyde-3-phosphate / TP / PGAL / G3P**;
16. **6** carbon dioxide molecules required for synthesis of **1** molecule of β-glucose;
17. **6n** molecules of carbon dioxide required for **1** molecule of cellulose where n is number of β-glucose in the molecule;

Max 3 pts

QwC: [1mark] Clear, organised flow without ambiguity for Calvin cycle description and to include **at least 1 description for at least 2 phases in the Calvin cycle + at least 1 mark for origin.**

- (b) In a particular plant species, which could produce offsprings in the hundreds, its height and flower colour are each controlled by a single gene locus. The two gene loci are located on different chromosomes. It is known that the alleles for tallness (T) and red flower (R) are dominant to the alleles for shortness (t) and white flower (r) respectively.

Using one molecular method and one genetic cross method, explain how the genotype of a tall, red-flowered plant of this species can be determined. [13]

Molecular method:

1. Ref **extraction** of genomic **DNA** from sample;
2. Forward and Reverse **primers** designed for **alleles t** and **r**; (A: primers for alleles T and R)
3. **Polymerase Chain Reaction** conducted using the designed primers **separately**;
4. Using thermostable DNA polymerase / Taq DNA polymerase;
5. **Ref. Denaturing** involving **heating** to **95°C**;
6. **Ref. Annealing** involving **cooling** to **55°C (A: 50 – 65°C)**;
7. **Ref. Elongation** involving **heating** to **72°C**;
8. Ref. cycle **repeated** 30-40 times for PCR;
9. **Gel electrophoresis** conducted;
10. Blotting process: DNA fragments (in the gel) are transferred onto **nitrocellulose paper/membrane**;
11. ref. nucleic acid hybridization - treatment with a **single-stranded radioactive DNA probe** which **binds/anneals to alleles t and r** via complementary base pairing;
12. Bands are visualized on X-ray film / autoradiography.
13. Presence of band for allele t indicates Tt genotype / heterozygous for gene locus for plant height while presence of band for allele r indicates Rr genotype / heterozygous for gene locus for flower colour;
14. absence of bands indicates plant is homozygous dominant at that gene locus;

OR

13. Presence of thicker bands for T allele indicates homozygosity (TT) + thicker bands for R allele indicates homozygosity (RR) / thinner bands indicates heterozygosity;

Max 8 for method + Max 2 for pt 13/14

Genetic cross method:

1. Perform a **test cross**;
2. If offsprings are all tall, plant is homozygous dominant at gene locus for height / ref. correct determination of TT genotype;
3. If offsprings are all red flowered, plant is homozygous dominant at gene locus for flower colour / ref. correct determination of RR genotype;
4. If short offspring present, indicates Tt genotype / heterozygous for gene locus for plant height;
5. if white flower offspring present, indicates Rr genotype / heterozygous for gene locus for flower colour;

(Accept: genetic cross diagrams for TTRR, TtRr, TTRr and TtRR for pt 16-19)

Max 5

QwC: [1mark] Clear, organised flow without ambiguity and to include 1 mark each for both methods and determination of results.

QUESTION 5

- (a) The Centers for Disease Control and Prevention in the United States of America recommends an annual influenza vaccination for everyone 6 months and older with rare exceptions, for example, those with life-threatening allergies to the vaccine. Each annual vaccine is different and contains three influenza strains.

Explain the need for a person to receive influenza vaccinations annually for protection against the disease and suggest why a person may still get the disease even after receiving his/her annual dose of vaccine. [12]

[need for vaccinations annually]

1. Ref. Different strains exist;
(New influenza virus strain can emerge via Antigenic drift and Antigenic shift)

[Antigenic drift]

2. Ref to **antigenic drift + Spontaneous mutations** in the viral genome coding for antigens haemagglutinin and neuraminidase,
3. due to a **lack of proof reading of viral RNA-dependent RNA polymerase**,
4. causing minor **changes in the 3D conformation of haemagglutinin or neuraminidase**;
5. **Ref. changes may accumulate to become a new strain**;

[Antigenic shift]

6. Ref to **antigenic shift + Two (or more) different strains** of the influenza virus infects the **same host cell**;
7. There is **reassortment** of the **viral RNA segments**,
8. giving rise to a new combinations of RNA segments in new viral particles, hence **new combinations of surface antigens haemagglutinin and neuraminidase arises**, a **new virus strain** results;
9. Annual vaccines contain only **3 strains** that are **common** for that **year / season** / Ref annual vaccine does not provide protection for all strains;

[suggest]

[infection by one strain only]

10. exposed to an influenza virus **shortly before getting vaccinated OR during the period that it takes the body to gain protection after getting vaccinated**.
This exposure may result in the person becoming ill with influenza before the vaccine begins to protect the person;

[infection by other strains]

11. ref. exposed to a flu virus that is not included in the seasonal flu vaccine;
12. Ref. influenza vaccine is made to protect against the three strains that research **suggests** will be most common / likelihood that prediction for 3 most common strain not being accurate;
13. Ref existing memory cells / antibodies do not recognise antigens on new strain;
14. Ref. Protection provided by vaccination can vary widely, based in part on health and age factors of the person getting vaccinated / ref. may not provide protection in people with **immune system that are not fully functioning or developed**;

QwC: [1mark] Clear, organised flow without ambiguity and to include at least 1 mark for each section.

- (b) Free nucleotides are present in various locations in a eukaryotic cell. They can exist both in the more commonly known non-cyclic form or in the cyclic form.

Describe the general structure of a non-cyclic form of nucleotide and with named examples, outline the role of free nucleotides, cyclic or non-cyclic, at particular locations in a human cell. [13]

[Describe structure of nucleotide]

1. A nucleotide consists of 3 components covalently bonded together: a nitrogenous base, a pentose sugar / 5-carbon sugar and a phosphate group;

[Nitrogenous base]

2. Adenine (A), **Thymine** (T), Cytosine (C), Guanine (G) in **DNA nucleotides** and Adenine (A), **Uracil** (U), Cytosine (C), Guanine (G) in **RNA nucleotides**;
3. Ring structure of each base can either be purine or pyrimidine;

[Pentose sugar]

4. Ref. deoxyribose in DNA nucleotide and ribose in RNA nucleotide / OH group on carbon-2 in ribose is replaced by hydrogen atom in deoxyribose;

[Phosphate group]

5. Ref. possibility of **monophosphate**, **diphosphate** and **triphosphate**;
6. The base is joined to **C-1** of pentose, the phosphate group is joined to **C-5** of pentose; (Accept: drawing showing the joining)

[role of free nucleotides]

7. **Deoxyribonucleotides/ DNA nucleotides** in nucleus used as raw materials for DNA replication to synthesise DNA molecules;
8. **Ribonucleotides / RNA nucleotides** in nucleus used as raw materials for transcription to synthesise rRNA, tRNA and mRNA;
9. **cAMP** generated from ATP in **cytosol/cytoplasm** in a reaction catalyzed by the active form of a membrane-bound enzyme known as adenyl cyclase acts as **second messenger**;
10. ref. **signal amplification** during **signal transduction** in cell signalling;
11. **GTP** in **cytoplasm/cytosol**, when bound to G protein, activates G protein / ref. inactivation when GDP is bound;
12. Ref. allows **switching on and off** of the **transduction pathway** during cell signalling;
13. **ATP / GTP** in a **named location** + **hydrolysed** to provide energy in a **named metabolic process**;

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14. ADP in mitochondrial matrix used for ATP synthesis by ATP synthase OR ADP in cytoplasm / mitochondrial matrix used for ATP synthesis in substrate level phosphorylation;
15. AVP; (eg. Distribution of various nucleotides across a membrane affects electrical gradient due to negative charges of phosphate groups)
16. AVP; (e.g. addition of 5' guanosine cap in the nucleus during post-transcriptional modification)

QwC: [1mark] Clear, organised flow without ambiguity and 1 point for nucleotide structure + 1 point for role of nucleotides